Data and Keywords Null Essay

Subheading and connectors

All that glitters is not gold

Uncovering the silver lining

finding hope in diversity

charting a path forward

Catchy Lines

Too much of information has created Poverty
of attention

Feminism isn't about making women stronger ,
they are already strong ; it's about changin
perception

GD Anderson

"When women lead, communities flourish." —
UN Women India

“Inclusion sometimes needs exclusion—to
uplift those who were always left out.”

Catchy Lines

GS1 India is a bundle of contradictions held
together by strong threads of pluralism

GS2

Ek Bharat Shresh Bharat

Sabka Sath, sabka vikas, sabka vishwas, sabka
prayas

Where voices are many policies are wise ,

Disasters test governance, not just geology

Health is not a luxury but a right - Amartya Sen

Good Governace is not uniform governance - it

is the governance that fits people Raghuram Rajan

Conclusions Empowerment begins when institutions act not
just advise

X , must become India's engine for Y

India's 5S which it always stood for

Samman

Samvad

Sahyog

Samriddhi

Samadhan

4Ds

Democracy

Digital tech

Diplomacy

Defense

The soul of India lives in its villages.

GS3

Mountain Criers before it falls

Air is not negotiable

Governance must flow where air/water do

Terror Feeds Crime and Crime Shields Terror

“Ending poverty is not charity, it is justice”

Atithi Devo Bhava

Shared Nature of Public Space

Intros

X is the foundation of Property and
Productivity or Progress yet India struggles
with its availability , accessibility, quality.

Conclusions

X is foundation for Human Development and
Inclusive growth thus India should make it
equitable and accessible to all...

GS 1

Society

Elder population By 2050 21% UNFPA

Cencus 2011

Demography

Age wise

0-14 31%

15-60 61%

60+ 8%

Median age

24.9 yr

China 37

Demographic deividend peak

2041

20-59 59%

PwD

2.2%

2.7 Cr

Tribals

10.45 Cr

8.6%

SC 16.6%

Trans 4.87 L

Literacy Rates

ST 59%

Women 65%

SC 66%

Overall 74%

Men 82%

Bihar 62%

Kerala 94%

Literacy Rates

Women 71%

Men 84%

Overall 77.7% NSO , 2023

NFHS-5

slums Population

5.4%

17% of urban population

Diversity and pluralism

Intro

Data

Language spoen > 10000 people 3

Religion 1

ST 1

ethnic groups 1

Examples

Religious

Tribal

West India Bhils

Arunachal Apatanis

differ in dress language and ritual

Geographic-cilisation variation

Buddhism in Himalayan

tamil shaivism

Tribal totem in Bastar

Composite Cutlure Sufi-Bhakti Tradition

Tolrerance

Onam celeberated by christian, hindus and
muslims

Issues

Marginaliation Tribals

Niyamgiri (Odisha) mining and rehablitation

Language

Bhili and Gondi speaking student lag behind in
Hindi speaking schools

NCERT learning outcome study

Language endangerment in India

197

Extinct In Last 5 decades 50 People's Linguistic survey

UNESCO

Cultural Homegnisation western clothing

Identity Politics Delhi Riots 2020

Family and Marriage

Family

Nuclear Households

58%

51.7% Cencus 2011

NFHS 5

Group of parents, children or couple living
together as a unit

Types

nuclear parents and unmarried children

joint multi gerneration

weekend Corporates

extended family NRI

Nature of Change

types extended, nuclear, weekend

nature

single parent

dual earner

reconstituted (divorced and remarriage)

Live in relationships

Issues in Family

Women

gender based divison of work women perform X% of domestic work 85-90% NSSO

violence 30%

dowry deaths 6800 NCRB 2022

Elders Neglect 71% HelpAge India Report 2021

Social control mechanism Khap Panchayats

Conclusion Structure to spirit

Marriage

Divorce Rates

India 1% of marriage

US ~ 35-40%

Types

Monogamy

Polygamy

Polygny one man married to more than one women Gonds

Polyandry one women more than one man 1

Intercaste

5-6%

in urban centers 10-12%

India Human Development Survey

arranged 93%

love

Inter religion 2.1%

Functions

legitise sexual relationship

expand kinship

reflects religious duty (Dharma)

Change in Nature

Delayed Marriage

rising divroces

reduction in #children

Delayed Procreation

State sponsored inter caste marriage

Women

Early Vedic Poets

Apala

Ghosh

Later Vedic Gargi debated with Yajnavalka Brihadaranyaka

colonial rules

Bengal Sati Regulation 1829

Bengal Child Marraige Regulation 1799 , 1804

Widow Remarriage Act 1856

Age of Consent Act 1891

Organisation

Pre Independence

Women India Association (WIA) (1917)

adult sufferage demand achieved 1930

Anne Besant + Margret Cousins

All India Women Conference (1927)

Girl Education

Set up Lady Hardinge College 1932

Margret Cousins

National Council for Women in India (1925) National branch of International Council

Post Independece

SEWA

provide livelihood , trade union in unorganised
sector

Ela Bhatt 1972

All India Democratic Women Association

(AIDWA)

Dowry, discrimination, violence

Bhartiya Grameen Mahila Sangh (BGMS) Organised women into SHG

Indian Association of women's studies (IAWS) academic gender discourse

Post Independece Movements

Telangana Movement against Landlords (1946-
51)

Chityala Ailamma

Anti Dowry Dahej Virodhi Chetna Manch (1980s) Dowry Prohibition Act 1981

Chipko Movement 1970s

Narmada Bachao Medha Patekar

Addidtional Data

Esther Duflo Nobel Laureate linking women
empowerment with investments in hyegine and
social welfare

Chhavi Rajawat

MBA women Sarapanch of Soda (rajasthan
village)

Solutions

RGSY

Zipper Method of Rwanda reserving every third seat in party for women

State Institutes of Rural Development Jharkhand

Secularism

Challenges Communal Tension Murshidabad Violence 2025

India vs west

sepration vs equidistance

uniformity vs religious plularism in laws (hindu
marriage act, muslim personal law)

state regulation of religion (sabrimala temple
entry)

state sponsorship in religion Hajj, Amarnath Yatra

Subtopic 5

Population and associated Issues

First Cencus 1872

National Population Policy

Maternal

MMR <100

institutional delivery 80%

trained person delivery 100%

TFR 2.1 by 2010

stable population 2045

Education free and compulsory till 14

Marriage after 20 in girls

Crude Birth rate

2012 21.6

2021 19.3

Crude Death Rate

1971 14.9

2020 6 SRS 2020

TFR

Bihar 3.0

National average 2.0

1992 3.4

IMR

2021 27

2006 57

Life expectancy

independence 32

Present 70

Migration

Impacts (internal)

Positive

remittances

better jobs and education

gender empowerment (girls moving to cities)

Cultural Exhange

Negative

old age loniliness

informal works of migrants

cultural marginalisation Bihari discrimnation in host states

slums 5.4%

Population Pressure

Impacts (Externals) Brain Drain

Govt Steps

one nation one ration card

e-shram registration

Meri Pehchaan Migrant ID

Inter State migrant worksmen Act

Ageing

by 2050 60+ 21%

rise in elderly population b/w 2011 - 2021 41%

SRS

SRS

GS2

Data Budget CSS

2015-16

2.04 L cr

11.5% of Budget

2025-26

5.05 L cr

10.5% of budget

Governance ···

Polity ···

SHG/NGOs ···

Vulnerable Groups ···

· · ·

Developmental Schemes in Health and
Education

IR

Rare Earth Resilient Supply Chain Initiative (RSCI)

India- Japan- Australia Pact to reduce
dependence on China for rare earth

EU India Trade and tech Council

To build alternative in clean tech and telecom
hardware

GS3

Data Bank

Economy

Inclusive Growth

Intro

UNDP : process and outcome in which all
section of society participate and benefit
equitably

Inclusive Developemement Index by WEF 2

Positive

2

% of Multim dimensionally poor household
[NITI AAyog]

HDI 4

Negative

Poverty still 11%

Out of Pocket Health Expenditure 47%

Pay gap in women among regular wage worker 24%

Education Pratham Report 1

Regional Disparity

southern states (4+telangana) contri in GDP 1

Per capita GDP difference 1

economic inequality

top 1% 2

Bottom 50% 15% wealth

Initiatives

Regional Disparity

Aspiration Districts 1

Aspirational Blocs 1

PM JANAM

Border Area Developement Programme

PMJAY 55 Cr Benefieciaries

PMGKY 2.2 L Cr

Sectoral Compositon

Total Workforce 60 Crore

WPR

Women 36%

Overall 56%

Men 76%

Labour forces share

Agriculture 45%

Service 30%

Manufacturing 25%

MSME 11.1 Cr

Budget

Money Comes From

Borrowing 24

Income Tax 22

GST 18

Corporation Tax 17

Money Goes to

States 22%

Interest 20%

Central Sector 16%

Finance Commission and other transfers /
Defense / CSS schemes

8%

Pensions

4%

China/USA 5%

Fiscal Deficit

2018-19 3.4

2020-21 9.2

2024-25 4.9

2025-26 4.4

PM

11.21 L crore ; 3.2% of GDP

2014-15 1.6%

Engines of Development

Agriculture

MSME

Limits 9

Credit Cards

Limt Rs. 5L

To be Issued 10L

State Finacnce

High Revenue Deficit State Punjab - 30%

States with debt to gdp > 35 10 in 2022 [RBI]

MSME

Significance 9

Challenges

Credit Penetration 14%

Indsutrial Land Regulation limits land usage unquantified cost

Govt Steps

Easy LIC Udhyam Portal

Credit

Emergency credit line Guarntee scheme

TreDs digital invoice discounting

Loan restructuring permitted by RBI for one
time in 2020-21

Solutions

changes in MSME Development Act. U. K. Sinha committee to bring

Credit

Credit Rating Agency for MSME

TurnAround time for bank to clear MSME loans 7-10 days

MSME Bond Market

Fund

Stressed Asset Fund 5000 Cr

Women centric MSME Fund

SIDBI transform into MSME development Bank

Cluster Base Approach

Easy exit and reentry option (insolvency)

Reduced LIC Cost

Capacity Building Amazon MSME Digitisation Intiative

No foriegn tender below for foriegn
procurement

200 Cr

PLI

Apple shift iPhone production to India 25% by 2025

India global electronic exports growth 28%

Started in 2020

sectors 14

Semiconductor

FPI

Pharmaceuticals

GST

Benefits

0% tax on exports

Increased Revenue GST Collection

2017-18 1L Cr

2023-24 1.6L Cr

Increased Household savings as Low GST on
essential as per [Mo Finance]

4%

Exapnsion of tax payer Base 60L 1.2 Cror

Negatives

Lack in enforcement of Anti Profeetering
Measures

April to Dec 2024 tax evasion as per Mo
Finance

1.88 L cr

Complex and confusion 3 diffrential rates on popcorns

Dispute and Litigation Mohit Mineral Case

Unemplyement

'predominantly a youth problem'

youth forms 83% of unempleyed population

educated youth share

2000 54%

2022 65%

Skill lag

75% of youth unable to send email with
attachement

Casual worker 90%

workers Paid below minimum wages as per ILO
2018

33%

India Employement Report by ILO

Poverty

Data

History

Poverty Alleviation Programme since 3rd FYP
(1961-66)

Before 1991 poverty reduction rate 0.44%

After 1991 poverty reduction rate 1.3%

Poverty Reductions (15-16 and 19-21) HCR and intensity [NITI Aayog]

People escaping Multidimesnional Poverty 13.5 Cr

Subtopic 7

Poverty Rate (World Bank 2022 report)

SC 33%

ST 43%

SDG 1.2 Reduce Poverty by Half

Govt Actions DAY-NRLM women mobilised into SHG 9.3 Cr

Hunger

India Rank GHI

2023 111/125

2024 105/127

India Score GHI

2000 ; 2008 ; 2015 ; 2024 38 ; 35 ; 29 ; 27

Categoy Severe

Mid Day meal scheme

primary level

Calorie 300

Protein 8-12 gm

upper primary

Calorie 700

Protein 20gm

Under Nutrition

Stunting

NFHS-4 (2015-16) 38

NFHS-5 (2019-21) 35

Wasting

NFHS-4 (2015-16) 21

NFHS-5 (2019-21) 19

under weight

NFHS-4 (2015-16) 35

NFHS-5 (2019-21) 32

Anemia

Child 67%

Women 57%

Gig Workforce

2020-21 77L 1.5%

2030 2.35 Cr

NITI Aayog

LPG Reforms

Reforms Privatisation

Sectors dereserrved

Prior 17

Later 2

PSU

Classification of PSUs mini/nav/maharatna

Volunteer Retiring

Disinvestments

Positives

Growth rate

1990-91 5%

8th FYP (1992-97)

Actual 6.8%

Projected 5.6%

2007-08 9.3%

2023-24 8.2%

2024-25 6.5%

2025-26 6.3%

Composition Share of GDP

Service 45 -> 55

Agriculture 30 -> 18

Forex $1 Bn -> $700 Bn

FDI $1.1 -> $81 Bn

Household Savings 23 -> 31% of GDP

Exports as % of GDP 7% -> 22%

Negatives

Skewed Growth

Agri decrease, industry no gain, service
increase

Inequlaity Top 1 % share in national income in 1990 6%

Infrastructure

Ports

Turn Around Time

India 0.9 days

USA 1.5 days

Logisitics performance index ranking india

2014 54

2023 38

LPI rank, in shipments

2014 44

2023 22

Amrit kaal Maritime vision

India's share in shipping

current 6%

Projeted 12%

Cargo Handling capacity

Current 2500 mt

target 10000 mt

Green port guidelines

LNG bukering station by 2030

Renewable energy 60% by 2030

EV chargin station

Issues

Undercapacity and congestion [UNCTAD 2023
]

85-90% capacity

Last mile connectivity rail freight sahre 27%

Dominance of 12 major ports sidelining minor 54%

Cargo

By volume 95%

By value 70%

Railway

Freight

Current Frieght share

27%

1950-51 85%

2030 target under National rail plan 45%

Daily 3.3 mT

Generates less carbon than road 80%

Global rank 4

Length 68000 Km

Passenger moved daily 2.4 Cr

Issues

Low train speeds

Freight 42 kmph

express

India 60kmph

China 250 Kmph

overaging rail track 35000 out of 38000

Accidents and derailments Balasored 2023

Done by govt

Electrification of broad gauge 96%

KAVACH

current deployement 1500Km

2030 targest 35000 km

DFC

Food Security Self sufficiency incereals

1970s

Total production in 2023-24 332 MT

Agricultre

Macro Data

% of GDP 18%

Growth rate over last 5 years 4.2%

% of total exports

11.7%

$53 Bn

% of workforce 45%

Horti/livestock/aquaculture

% of GVA

33

30

2014-15 24%

2022-23 30%

6.7

CAGR (Decadal)

2.82% (Production wise)

7.13%

6.75%

% of Agri exports

7.5

9.3

15

Employement

-

21 Mn

28 Mn

Mission

Horti

Mission for Integrated Development of
Horticulture (MIDH)

National Horticulture Board (NHB)

Livestock

NDDB

Rashtriya Gokul Mission

National Livestock Mission

Aquaculture

PM Matsya Sampada Yojna

Fisheries and Aquaculture Infrastructure
Development Fund (FIDF)

Concerns

Horti

Climate change, water scarcity, post-harvest
losses

Livestock

Greenhouse gas emissions, resource depletion,
low productivity

Aquacultre

Environmental impact (mangrove loss), water
pollution, disease outbreaks

Food Processing

Intro

We export what we should eat and waste what
we shoudl Process

Conclusion

Backbone of AtmaNirbhar krishi, potential to
convert farms into internation food chains and
reduce rural distress.

GDP Contri

Agriculture 11.5%

Manufacturing 10%

CAGR (2015-16 - 2022-23)

5.35%

1.61 -> 1.92 L Cr.

MOSPI

% of total agri exports

23%

Rising CAGR @13%

13 Bn in 2022-23

% of total organised employement

12%

70L

Investment growth in last 7 year 7.25%

Budget

National Institute of Food Technology,
Entrepreneurship and Management to be set
up in Bihar

Post Harvest Losses in Perishables 20%

Issues

Low Level of Processing

India 10% of agri produce

Developing countries 60-80%

Fragmented supply chains lack of storage 37 mt

lack of processing infra

testing labs, effluent treatment plants, modern
processing units, etc.

Credit access 6% of MSME in food processing recive it

Lack of Skilled Manpower 80% unskilled/ informal NSDC , 2023

Red Tapism

Govt Stance

PM-Kisan sampada Yojna

Cold storage, food parks, etc.

Investment 10,000 Cr

Jobs generated 6L

PLI

in food processing sector (Mozarella Cheese)

In

Ready to eat

Millet

Low GST 0/5%

FDI 100% thtough automatic route

Operation Green

Originally for TOP

Now expanded to (TOTAL) 22 perishable items

Price Stabilisation

ODOP

e-NAM

Agri Infra Fund

1L Crore

Farm gate Infra

Primary Processing

Need to be Done

Agri-export zone

Cluster Developement JackFruit Cluster Kerala

Credit acess

Strengthen FPO and contract farming

Skill Development under Skill India Mission

Food Safety

Packaging

Transport and Warehouse

furtis and veggies transport via roadways 97%

Requirement 70 mT

Availability 35 mT

Shanta Kumar Committe Report

MSP & Market

Recommended 22

Covered 24

Benefits goes to 6% farmers Shanta Kumar Committe Report

Post harvest losses 15-25% ICAR

Land Reforms

Landless Households [Socio Economic Caste Census 2011] 30%

Negative

%person own less than 2 hectare [NSSO] 90%

agri cencus 2018

Landless 7

Marginal 75

Small 10

Positive Small and marginal owned area

1961 20%

2013 53%

Implementation

Surplus Above ceiling 72L Acre

Redistributed 21L acre

Good Implementation

Kerala (EMS Nambodiriprasha)

WB (CPIM)

MoRD

Farmer Debt

% of household spend more than they earn 70%

% in debt due to rise in farm input cost 50%

NSSO data

Irrigation

land under irrigation of Gross Cropped Area

2016 49%

2021 55%

NITI Aayog

rice and sugarcane take 60% of water available for irrigation

Rainfed Agriculture % of Prduction

Millets 90%

Oil Seeds 80%

Livestock/Cotton 60%

%of irrigated area under groundwater irrigation 60%

%of ground water usage for irrigation 90%

% of global fresh water resource in India 4%

Solution

Microirrigation can save upto [Ashok Dalwai
Committee]

40% of water

Major cause of water inefficiency [NITI Aayog]

Electricity subsidy

Use Metered electricity

Fertilizer

35% dependency

$9 Bn

Cropping Intensity

1950 111%

2019-20 151%

Area Under Food Grain (Cross Cropped Area)

1980 73%

2022 62%

Coarse cereals Area

1980 22%

2022 9%

Oil Seed

1980 9%

2022 15%

HortiCulture

2001 8.5%

2022 14%

MoAFW

Crops and produce

Cereal

2023 332 mt

1947 50 mT

Expected rise this year

6.6%

354 mt

Productivity

Wheat 3.4-3.6 Ton

Rice 2.8 Ton

India ~3 tonne / hectare

China ~6.1 tonne/ hectare

FAO

Millets

Demand per capita per month in rural india

1962 4.4Kg

1983 3.3 Kg

2011 0.4Kg

90% decline from pre green reovlution

Export 2nd Largest in world

Productivity 1.9 tonne per hectare

Rise as IYoM capagain 40% growth in sales Amazon

Climate Resilence <30 cm rain

staple crop for tribals 60% SECC

Benefits Health

Gluten free ;

high nutrients

Ragi 10x calcium than wheat or Rice

Bajra Iron

low glycemic index

Anti-oxidants

Conclusion Poor Man Grain -> Smart Food

MoAFW

Horticulture

352 mt 2023

2nd largest in world Production

pick 4 vital stats from above + change in area

Why a good sector

High income 5-10x in apple relative to wheat

Labour Intensive planting, harvesting, etc.

Nutritional Security Hidden Hunger

Exports $3.8 Bn

Challenges

Low Processing

2-3% in fruits

Global 60%

Post Harvest Loses 20%

Fragmented Supply Chain

Price Volatility

Credit Flow <5% of total agri credit

Climate Change Vulnearabilty

Govt Stance

MIDH (subsumes NHM)

production

infra

market

National Horticulture Board

Financial assistance for cold storage

Nurseris

Operation Greens : TOP -> TOTAL (incudes
fruits and vegetables)

subsidies on transport and storage

Horticulture CDP

Way Forward

CDP Kerala CDP Jackfruit

FPO promotion Banana in Jalgaon

Digitisation

Pest forecasting

AI Advisory

Kisan Sarthi App

Export oriented porduction Grapes/Litchis

Cold Storage through solar power

Crop Insurance

Pulses

Protein by weight 20-25%

Largest

Producer 25%

Consumer 27%

Importer 14%

Import dependence 25%

Solution:

Integrated Schemes of Oilseeds, Pulses, Oil
palm and Maize or ISOPOM,

Oil seeds Import Dependence 55%

Economic Survey 2024

Technology

AI

Issuess

Data Quality and Avalibilty Indian Data not in machine readble format 80% NASSCOM , 2023

Biases

High error rates in facial recongition of black
people

MIT Media Lab , 2020

Accountability Issues

Usage in Deepfakes and misinformation During 2024 elections

Job

Losses by 2027 8.3 Cr NASSCOM , 2023

AI Skill Penetration 3x global avg

Indian short on AI talent 2.9 Cr Linkedin Skill India Report

India's workforce at risk of displacement from
AI

12% IMF

Performing AI clusters

India 20

China 1800

Digital Literacy among adults 38% Meity 2023

Internet Penetration in Rural Area / Percentage
of internet users in rural india

55% TRAI

Applications

Agriculture

Remote sensing and real time crop data IIRS Dehradun

Prediction of crop outputs and disease Microsoft AI Sowing App

Education

LEAD School platfrom for personalisse
upskilling

Healthcare

Wadhwani AI in TB detection

INAI

Intel + Govt of Telangana usage of AI in public
healthcare

Security

AI faciial reconition

Drone AI surviellence

Infra and smart cities Traffic management in Surat and Bhopal

Governance SUVAS

Translation of Judgements by SC in regional
languages

India Govt done

India AI mission

10,300 Cr.

Compute, infra, dataset, startups

National Data Governacne Framwork To standardize anonymous data

INDIAai cetral hub for AI related devs in India

Bhasini Platform

Global Platform for AI (GPAI)

Internet

Data

95+ Crore

Price Drop since 2016 90%

CAGR since 2014 14%

Villages having access to internet 95%

Digital Literacy in Adults 38%

Smartphones

SmartPhone Devices in India

120 Cr+ (Mobile Phones)

Rural > 15 own

Female 75%

Male 79%

Urban > 15 own

Female 86%

Male 89%

Households with smartphone

In LWE 18% NFHS

In India 85% MoSPI

Programme BharatNet

Connect All Gram Panchayats with optical
Fibre Cable

Patent

File 2 nd globally in India

Commercialise 15% only

Environment

Climate Change

Data:

Related to climate

Rise in extreme weather events as per WMO 5 times in last 50 year

Rise in Average Temperature since 1901 0.7 IMD

Expected rise in India temperature by end of
decade

2.4-4.4 C IPCC AR6

Economimc Loss 2L Crore NIDM

Emmissions

India

3.3 Gt

3% global

3rd largest emmitter

Per capita

US 14t

India <2t

Climate Change Vulnerability top 6 Germanwatch

Gaps

Power generation from Coal 70%

Adaptaion funding is low 20%

Insufficient Green Finance for transtion by 2030 $2 Tn NITI Aayog

Govt stance

National Level

India's Panchamarit

Renewables

Current : 46.3%

50%

Non Fossil

500 GW

Current : 220 GW Solar 110

Rooftop Solar 17 GW

Ground 80 GW

Reduction in C-Emssions 1 Bn Tonne

C-Intensity Reduction Base [2005]

45%

Current : 36%

Net zero by 2070

National Action Plan For CC

8 subplans current expansion to 13 under
review

India Cooling Action Plan Year : 2038

sustainable cooling by 2038

Reduction cooling demand till 2038 20-25%

National Adaptation Fund for CC Rs. disbursed so far 1200 Cr.

LiFE Behavioral Change

International Level Founding Member of

ISA 110+ countries

CDRI 35+

Green H2 Mission

Production 5mmT

Jobs 6L

Investments 8L Crore

SIGHT for Electrolyser

R&D Hyddrogen Vallen for academia and industries

Transport

Ethanol Blend 20% by 2025

Share of railway frieght

Current 27%

by 2030 45%

Scheme

DRIVE

PM-eBus Sewa

FAME

Rise in EVs Expected to rise at CAGR from 2024-30 25-30%

Pollution

Intro

X Is a Public Health Emergency causing human
, economic, and envtl loss.

Air

Intro

Public Health emergency causing economic,
human and ecological damage.

Data

Support data of problem

Number of cities among most polluted cities of
India

9/10 IQAir, 2024

AQI Delhi in dec jan 450+

PM 2.5

in Delhi in Dec Jan 8 times WHO standard

contri of Transport 38% TERI-SAFAR

Causes

Stubble Burning in 2023 55000+ Cases ICAR

contri of Transport 38% TERI-SAFAR

SO2 emmission 70%

Construction and Road Dust contri in urban
particulate load

15-20%

Indoor Pollution 12 Cr. household

Impact

Loss of lives due to air pollution 16 Lakh Lancet

Reduction in Life Expectancy 5.3 Year Air Quality Life Index

Economic Cost 3% of GDP

Lost labour

Increased Healthcare

World Bank

What Govt is doing

NCAP

To Reduce Particulate matter by 2026 40%

Number of cities 131

CAQM For Delhi NCR and GRAP implementation

Air Act , EPA Act

EV FAME II

BS-VI fuel

Monitoring SAFAR System in metros

Case Studies

Delhi New vehicle that are EV 17%

Punjab Happy Seeder Subsidy 25%

Issues Poor Implementation of NCAP CAG report 2023

Conclusion

Air is not negotiable India's development
trajctory shall be written in clean and
breathable air

Water

Data in support

Polluted River Stretches 351 CPCB

Surface Water contaminated beyond safe limits 76%

Untreated seawage 37% MoHUA

Groundwater

Demand met in rural area by Groundwater 85% FAO

Rural Household relying on untreated ground
water

76%

Rainwater Hrvesting 11% CWC (Central water commission)

Over exploited unit 63% CGWB

Districts under water stress 234 NITI aayog

Water requirement for paddy 5000/kg CACP

Jal Jeevan Mission Report

CPCB

Impact Economic 6600+ Cr. in water borne disease MoHFW

Govt Initiative

Namami Gange STP

Jal Jeevan Mission

100% tap to rural households

Currently 76%

AMRUT

Focus on wastewater reuse and treatment in
urban

Rain Water Harvesting Catch the Rain

Conclusion

“Water pollution is not a pipe problem — it’s a
planning problem” — governance must now
flow where the rivers do.

Biodiversity

Hotspots in India 4/36

One of the megadiverse 17 countries

Area 2.4%

Representaiton in biodiversity 8%

Diaster Management

Intro

India has shifted from Relief Centric to Risk
Reduction Paradigm

Data

India's rank in Climate Change Vulnearblity 6th GermanWatch

Area prone to diaster 70% NDMA

GDP loss due to diaster 2% NIDM

People displace due to diaster in India b/w
2008-22

1.2Cr IDMC (Intl. Displacement Monitoring Center)

Public Buildings Following Diaster Resilient
Codes

18% CAG , 2023

Body

Legilative Framework

DMA, 2005

NDMA Under PM

NDRF MHA

SDRF States

Sendai for Diaster Risk Reduction 2015-30

Govt recent steps

Sachet and Damini Apps for Cyclone Alerts IMD

Apada Mitra

Trained village community voluniteers in 350+
districts

Vayulink (IAF) For radio communication in NE

Possible Diaster related questions

Landslides

Intro

Most common diaster in Himalayas , Western
Ghats region causing human, infrastructural
and ecological loss

Data

% area landslide prone 12 GSI , 2024

Examples

Waynad

Himachal

Shirur

Causes

Natural Eq , GLOF , fragile ecology , rainfall

Manmade

mining and quarrying Idduki and Waynad

unplanned hill cutting Chardham

lack of Slope stability checks Mid slope hotels and hydropower dams

Conclusion

Landslides are not just geological, but
governance challenges — integrating science,
community, and policy is the only sustainable
path.

Mountain Criers Before it falls

Earthquake

Intro

Eq are among the most devastating sudden
onset diaster exposing fault lines in diaster
governance and infrastructure resilience.

Data

Area of India under siesmic zone III to V 60%

Examples

Nepal Bihar 6.4 M

Assam 5.6 M

Body Vulnearbility factors

Unplanned urbanisation Shimla

lack of infra resilience

% of building in zone IV-V with seisim
meassures

95%

high population density in seismic zones Delhi

Tectonic Location

Floods

Intro

Most frequent diaster in India causing human ,
infrastructal and ecology loss

Data

Losses due to Flood 7000 Cr NIDM estimate

People affected annually by monsoon floods 70-80L MHA

Body

Types

Flash Chamoi

Urban Bengaluru

coastal odisha and wb

riverine Brahmputra

GLOF sikkim

Causes

Natural Causes

Human

Glof, excess monsson , cloud bursts, cyclones

encroachments of river bed(delhi), irregular
dam flow (bhakra) , unplanned urbanisation
(bengaluru) , siltation

Govt Stance

Flood watch app

FLOODS (2020) ISRO satteline flood mapping

Conclusion

India cannot control the monsoon, but it can
control its impact — through planning,
foresight, and floodplain governance.

Heatwaves

Intro

Heatwaves are becoming more frequent,
intense and longer in India, posing threat to
agriculture,public health and urban
sustainabilty

Data

Nuber of heatwaves days

2023 98

2024 170

Deaths [2024] Official 150+

hottest year in last 122 years 2024 IMD

States with Heat Action Plans 23/28 PIB

Govt stance

Ahemadabad first city for HAP reduction in
mortality

30-40%

India Cooling action plan (2019) sustainable cooling by 2038

Way forward

HAP Formed, adequate budget

social protection

cool public space, staggered works hours,
water accessibility

Infra

Green roof, vertical parks, green cover

Susutainable cooling

Community SHG, PRI to distribute ORS

Conclusion

Heatwaves are no longer future threats — they
are daily emergencies. The time to invest in
cooling justice and resilience is now.

Security

Naxalism

Intro

Most persistent internal security challenge due
to governance failure, alienation of property
and underdevelopment

Conclusion

By bridging governance, justice and local
empowerment India can tranform LWE Zone
into Peace Zones.

Body

Data

Incidents of violence

2009 2258

2023 509

Number of Deaths (civilian + security)

2010 1005

2023 92

Districts under Security Related Expenditure
Scheme

2010 90+

2045 45

Surrendere b/w 2016-2023 under rehabilitation
programme

7028

MHA

Causes

Land Alienation

imporper implementation of FRA, 2006 leaving
tribals landless

25L+ FRA claim rejected MoTribal Affair

FRA implementation status reports

UnderDevelopment

NITI Aayog SDG India index Ranks Lowest

LWE villages lack all weather road 70%

Police Attrocities Sardapur (Bijapur) fake encounter case

Exploitation of tribal grievance Janatana Sarkars (Parallel administration)

Govt steps

Security

SAMADHAN Doctrine (2017)

smart ops, tech use, motivation , intelligence
based policing

Security Related Expenditure Scheme

CRPF force, CoBRA units, GreyHounds in
Andhra

Bastar Fighters Tribal Youth into Police

Development

Aspiration Districts and Blocs Programme

Road Connectivity Project for LWE area

Rehabilitation

2.5L Package

Vocational training

Current Challenge

Human Right Violations NHRC Report

Weak PRI in LWE region

Digital Gap Households with smartphones 18% NFHS-5

MoPR 2023

Next Steps

Local Policing JanMaithri Kerala Model

Tribal Cadre in LWE regions NITI Aayog

Education Navodaya, hostels, vocational hub

Cyber Security

Intro

"Data is the new oil and Cyber threats are the
new Battlefield" , cyber security is no longer
optional but existential

Conclusion

As India Aims $1Tn Digital Economy Cyber
Security can't no longer be seen as cost but
core infra.

Body

Data

Rise in Cyber Crimes in 2022

24%

65000 FIR

NCRB

Incidents faced in 2023 13.9L Cert In

Rise in Ransomware Attacks in 2023 53% KasperSky

Indian rank globally in phising attacks 4th IBM

Impact Financial Loss in 2023 1714 Cr RBI

Major Threats

UPI and finance

Sim Swaps

Phising

Fake Loan Apps

State Sponsored

Chinese Group targetting Ladakh Power grid in
2023

APT41

Attact on Covid-19 Vaccine by Korean Group Lazarus

Ransomware AIIMS Delhi 2022

Social Media Manipulation Deepfakes in 2024 elections

Govt Stance

Law

IT Act Main Law for cybercrim and e-gov

DPDP Act for data protection

Agencies

Cert In Preimer Agency 24*7

NCCC National Cyber Coordination Center Monitors internet traffic

I4C

Coordinates among 100+ cyber crime units in
states

NCIIPC (National Critical Information Infra.
Protection Center)

to protect power grid, telecom, finance , etc.

Social CyberCrime Volunteer Framework Engage citizen to report unlawful content

Issues

Fragmented response

State Police

Cert-In

I4C

Outdated IT act No provision for deepfakes, AI, Ransomware

Lack of talent

Required 10L+

Current 1

NCRB

Under reporting 25%

Conviction rate 2%

Weak MSME Protection % not using any formal cyber hygiene 82%

NASSCOM

To Do Under Skill India Mission Cyber Sikhsa 2.0

Border Area Management

Intro

Long Border of 15000 Km , along 7 countries ,
with tough terrains including deserts, himalyas,
and dense Jungles.

Conclusion

A strong border is not build upon barbed wire
and fence alone but with empowerment and
inclusive development.

Body

Data

Border Length

15106 KM

With Bangla 4096

China 3488 KM

Pak 3323 Km

Border Area Development Programme 16 states are covered

Issues

Security and strategic

Ceasfire violation, Chinese Transgression , UAV
intrusions, State terrorsists,

Developmental

Underdeveloped

Poverty, tribal, lack of roads and connectivity,
telecom connectivity

Govt Stance

Vibrant Village Programme for borders along
China

Road, 4g , solar power,

Roads

India China Border Roads

Roads targetted by 2025 73

Length 4203 Km

Tunnels Atal, Zojila, Sela

Tech

CIBMS (Comprehensive integerated border
management system)

Radar, thermal sensors, UAVs

use AI facial recongition, smart fences,
sateelite feed.

Security mdernisation Night vision, encrypted comms, drones

Issues

Many Border regions are governed by army
personnel due to lack of IAS IPS

Topography and bad weather

Drugtrafficikking via west and east

Militancy Kuki-zo Militant trafficking via Myanmar

Migration Rohingya

Alienation and local discontent Manipur riots (ongoing)

Solutions

IAS Cadre + Extra Renumeration

Flood Lights

Panchyats integeration into VVP

MGNREGS+BRO for empowerment and
development

Organised Crime and Terrorism

Intro

Organised Crime is the oxygen of Terrorism , as
criminal syndicates provides arms, finance,
and logistical support to terror networks.

Conclusion

Crime Feeds Terror and Terror shields crime,
India must dismantle both arms with speed
and synergy.

Body Data

Linkage LeT, JeM linked to

Hawala

Drug trafficking

Arms trafficking

BSF Shot drones carrying AK-47 and RDX in
Amritsar 2023

NIA Annual Report

India Rank in Global Terrorism Index 13 IEP (institute of Economics and Peace)

Narco-terrorism

3000Kg Heroine from Gujrat busted linked to
Pak based terror Syndicate

NIA

Arms nd explosive

which Company operatives provided arms for
1993 Mumbai Blasts

D-Company

Drone based arms drop increased by 300% BSF , 2024

Cyber attacks + fake crypto funds ISIS uses CERT-IN

Issues State v/s cnter

Organised Crime State

Terrorims Central

Money Laundering

Concelaing the origin of money

Process PLI

Placment

Layering

Integeration

Cost to GDP 2-5%

Knowledge Diplomacy

Student

Exchange

Study abroad

Talent

academic

professional

University

MoU

campus in foreign countries

GS4

Examples of topics

Laws

Law help in ethics

CSR under companies Act 2013 Philanthropic ethics

Environement Protection Act , 1986 environmental ethics

Outdated Prisons Act , 1894

Void Male sexual harassment

In conflict with morality

Marital Rapes

Subtopic 2

Vary with jurisdiction Western countries not allowing abortion

Conscience

Crisis

Professional duty v/s religious freedom

Doctor who considers abortion is a sin, need to
perform legal abortion

Benifits social reforms

Aboliton of Sati/Untochability

Decriminalisation of section 377

(Homosexuality)

Law/rule vs consicience

Accountability E-governace

DBT

CPGRAMS

Environmental Ethcis terms

Sustainabilty

Intergenerrational equity

Common but diffrentaited responsibilty

Distributive justice Climate Finance

Corporate Governance Uday Kotak Panel 2017

Misuse of Data Cambridge Analytica Scandal

Uninformed data collection of millions usage
by political Consultancy

Misuse of Power

Sardapur fake encounter case erodes trust

Custodial violence victim father son duo in
karrnataka

DM surajpur slapping Youth for violation of
curbs during covid

Main Topic 3 Main Topic 4 Main Topic 6 Main Topic 10